

Alex Cunningham PhD – Role Profile

Main duties and responsibilities	
1	SOFTWARE PROJECT MANAGEMENT
	The role holder leads the development of CanRisk Web GUIs and Web services. These activities encompass all aspects of the software development life cycle. Key responsibilities are set out below: • Lead the development of Web GUIs and Web services and assume responsibility for the delivery of corresponding software products within an acceptable timeframe • Act as line manager to the BOADICEA Team Web developers, supervise their work, agree targets and draft workplans • Guide developers as they translate software requirements into prototypes/designs and prototypes/designs into working software • Guide developers as they address practical problems of software implementation, Web security, and data protection in a clinical setting • Interpret guidance on software engineering best practice • Lead efforts to adopt modern software engineering tools (e.g. GitHub, automated testing, JIRA, Confluence) and modern implementation technologies (e.g. Python, Django) • Conceive, design and develop technologies to widen access to breast/ovarian cancer risk models further (e.g. Web services) • Facilitate the capture of developers' expertise (e.g. via JIRA Service Desk and the Confluence knowledge base) to make the BOADICEA Team more robust • Communicate software project progress to PIs in monthly meetings
2	REGULATORY AFFAIRS
	The role holder manages the regulatory affairs (RA) work required in order for software products to be used in clinical practice. Key responsibilities are set out below: • Lead the work to deliver technical documentation required for regulatory approval in the UK/EEA and other jurisdictions (for EU IVDD, EU IVDR, UKCA etc) e.g. technical files, software process files, software quality product/process standards • Setup and manage meetings with RA consultants, precis and present findings to PIs for their consideration • Appoint RA consultants to guide the team through the regulatory process and work closely with these individuals to overcome problems that arise • Establish the qualification of the medical device (MD vs IVD MD) • Establish the classification of the medical device (Class I vs IIa) • Interpret manufacturing standards (e.g. EN 13485, EN 14971, EN 62304, EN 63266) and legislative documents (e.g. EU MEDDEV 2.1/6) and apply them as necessary to the regulatory process • Implement and maintain an EN 13485 quality system with RA consultants

	• Interpret guidance on software engineering best practice (e.g. software quality management, software risk management) and apply it to the regulatory process • Interpret guidance on GDPR and apply it to software designs and the regulatory process • Develop workplans, and work with the team to overcome the substantial barriers to the regulatory process that exist in the university e.g. the apparent lack of relevant in-house expertise or supporting infrastructure • Communicate regulatory project progress to PIs in monthly meetings
3	LIAISON AND REPORTING
	The role holder reports project progress to BOADICEA/CanRisk PIs in monthly meetings. The role holder also liaises with: • BOADICEA developers to manage their work • BOADICEA statisticians on aspects risk model behaviour • CanRisk stakeholders (healthcare professionals) on aspects of software design • RA consultants and the CRUK Early Detection Programme Technology Translation Manager to facilitate the regulatory process, translation and commercialisation • Research collaborators on software issues, regulatory issues, and scientific papers • University Research Operations Office on manufacturer registration with the UK competent authority (MHRA) and the MHRA itself • University Legal Services Office on software license agreements and GDPR privacy policies • BOADICEA/CanRisk end users to assist with their inquiries • Cambridge Enterprise to facilitate commercialisation of software products • NHS Digital, NHS Business Partners and commercial software companies to facilitate integration of software products with NHS and other computer systems • CanRisk qualitative researcher/social scientist on aspects of software design and evaluation
4	APPLICATION SUPPORT
	The role holder takes overall responsibility for supporting BWA v3 and BWA v4 <i>Beta</i>, legacy software products used by thousands of healthcare professionals; and guides developers as they support the CanRisk Tool. Key responsibilities are set out below: • Deal with BOADICEA/CanRisk Tool user inquiries • Maintain the legacy BOADICEA WordPress Web site • Maintain the ccge-web2 virtual server LAMP stack • Contribute to online documentation and FAQs held in the CanRisk Knowledgebase • Maintain the BOADICEA mailing lists
5	DATABASE ADMINISTRATION
	The role holder has setup and maintains a relational database (MySQL/Linux) to monitor software usage. This involves designing/refining database schemas,

	implementing the corresponding relations, modelling functional dependencies between attributes, writing queries, eliminating update anomalies etc.
6	PRESENTATIONS AND PUBLICATIONS
	The role holder gives oral/poster presentations at meetings and writes and contributes to scientific papers.